

Guruprerana Shabadi

PhD Student in Computer Science
University of Pennsylvania

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Education

- **University of Pennsylvania** Philadelphia, US
PhD, Computer Science. Advised by Rajeev Alur. Aug. 2024 —
- **École Polytechnique and Université de Paris Cité** Paris, France
M.S., Computer Science (MPRI) Sep. 2022 - Aug. 2024
- **École Polytechnique** Palaiseau, France
B.S., Mathematics and Computer Science, Summa cum laude Sep. 2019 - Jun. 2022

Publications

- Aaditya Naik*, **Guruprerana Shabadi***, Rajeev Alur, Mayur Naik. Do We Need Frontier Models to Verify Mathematical Proofs? *Third Conference on Language Modeling (COLM), 2026.*
- **Guruprerana Shabadi**, Rajeev Alur. Risk-Sensitive Agent Compositions. *International Conference on Learning Representations (ICLR), 2026.*
- S Hitarth, Alessio Mansutti, **Guruprerana Shabadi**. Optimization Modulo Integer Linear-Exponential Programs. *ACM-SIAM Symposium on Discrete Algorithms (SODA), 2026.*

Preprints

- **Guruprerana Shabadi**, Kaushik Mallik.
Auction-Based Online Policy Adaptation for Evolving Objectives. *Preprint.*

Workshop Papers

- **Guruprerana Shabadi**, Nathanaël Fijalkow, Théo Matricon. Programmatic Reinforcement Learning: Navigating Gridworlds. *Generalization in Planning Workshop, AAAI 2025.*
- Luca Negrini, **Guruprerana Shabadi**, Caterina Urban. Static Analysis of Data Transformations in Jupyter Notebooks. *SOAP, ACM SIGPLAN Conference on Programming Language Design and Implementation, 2023.*

Awards

Global Undergraduate Awards - Global Winner in Computer Science (recognized for work on <i>Static Analysis of Data Science Notebooks</i>)	Sep. 2022
Labex DigiCosme Scholarship	2022-2023
Excellence Scholarship from the École Polytechnique Foundation	2019-2022
DGA Scholarship from the French Directorate General of Armaments	2021-2022

Research Internships

- **IMDEA Software Institute** Madrid, Spain
Research Intern Apr. 2024 - Aug. 2024
 - Advisor: Alessio Mansutti
 - Topic: Integer linear-exponential programming
- **Institute of Informatics, University of Warsaw** Warsaw, Poland
Research Intern Apr. 2023 - Aug. 2023
 - Advisor: Nathanaël Fijalkow (Automata Theory Group)
 - Topic: Programmatic reinforcement learning
- **INRIA** Paris, France
Research Intern Jan. 2022 - Mar. 2022
 - Advisor: Caterina Urban (ANTIQUE Team)
 - Topic: Static analysis of data science notebooks using abstract interpretation
- **Department of Computer Science, École Polytechnique (LIX)** Palaiseau, France
Research Intern Jun. 2021 - Dec. 2021
 - Advisor: Sergio Mover (Cosynus Team)
 - Topic: Accounting for communication delays arising in cyber-physical systems